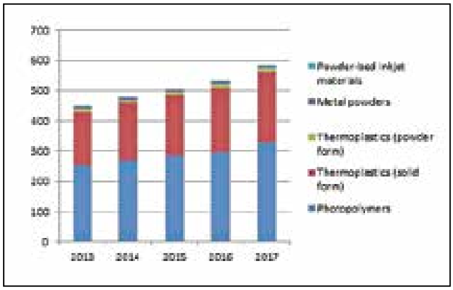
And then you have the more resilient and traditional manufacturing materials such as metals and complex polymer. Sturdier additives such as different types of steels, titanium, precious metals such as gold and silver, and other metals are currently not easily available or used in consumer level 3D printers. However, research and interest in their use are helping them slowly work their way outside of scientific grade 3D printers into consumer grade printers.



Polymers will continue being the top choice amongst the types of additives used

The potential of a variety of other materials such as nylon, glass-based polyamide, epoxy, wax and photopolymers has not gone unnoticed either and they're also finding root in the world of 3D printing. In fact, just about any material (besides a few exceptions) can be used as an additive. It's just a matter of incorporating the materials into existing or emerging printer designs.

- A few examples of ways in which exotic and unique materials are being used in 3D printing:
- Chocolate additives used to make custom designed desserts using CAD applications
- Bio-ink or stem cells to print human tissues such as blood vessels, bladders and kidney parts for surgery
- Silicon, calcium and zinc as additives to make an artificial bone structure on which real bone tissue could grow

Types of 3D printer technologies

Now that you know about the process of printing and materials used, it's time for an overview of the different types of 3D printer technologies and ways in which they differ. The methods they all use are of an ever-evolving nature and as newer techniques are perfected, they're bound to undergo change. For now, 3D printers are broadly based on four operational methods.

1. Traditional 3D Printers

Traditional 3D printers employ the simplest method of them all. The addictives are layered in a basic two-dimensional way across an X-Y